02 November 2026 14:45

What is machine learning and WHEN TO USE IT
As an FDE, ML Engineer…..

ML should be your last choice!

First the choice should be rules!

Create a deterministic system:
1. Rule no.1
2. Rule no.2
3. Rule no.3
4. ……..

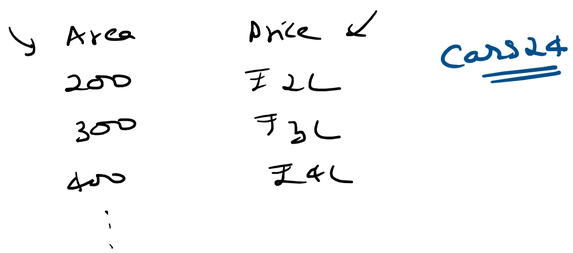
Only if the rule base it too large to create, then only go towards ML.
Using ML, computers figure out the rules on its own.

FDE, Paramount Cables.
That every month they need to allocate orders on machines.

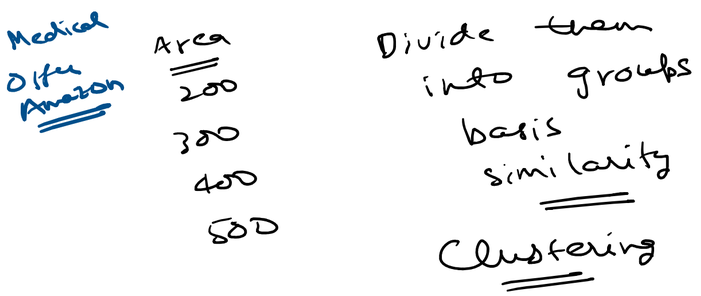
Every machine a specific capacity based on type of wire being produced.
Machines are also different for different materials
Client priority.

Rule Based System.

ML -> 4 categories
1. Supervised -> You know what to predict, there is input + output

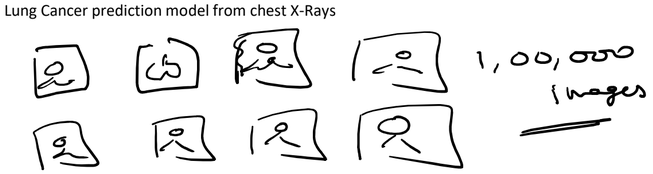


2. Unsupervised -> Where only input is available, nothing to predict



3. Semi Supervised

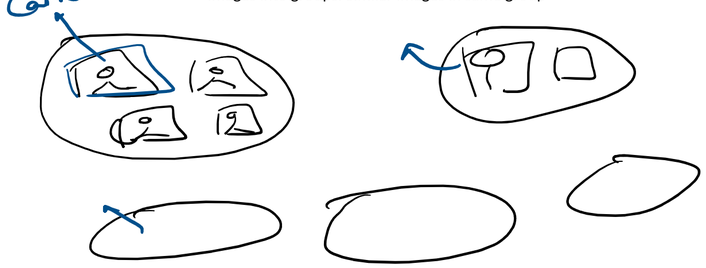
Lung Cancer prediction model from chest X-Rays



1. Unsupervised learning

Cancer

Images into groups. Similar images in same group



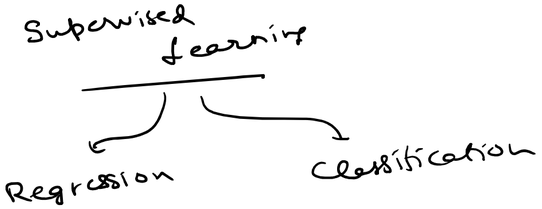
4. Reinforcement Learning

Model can prepare its dataset also on its own and train itself.

Chess -> Humans learnt chess
Old patterns, 1980, move -> then this happened. Playing more games.

Two machines -> Playing against each other.

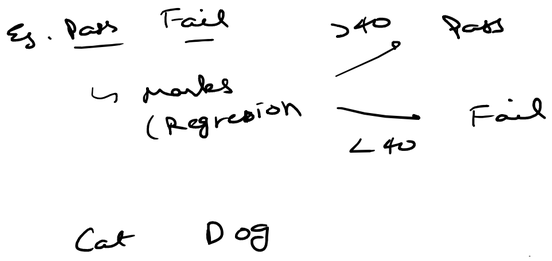


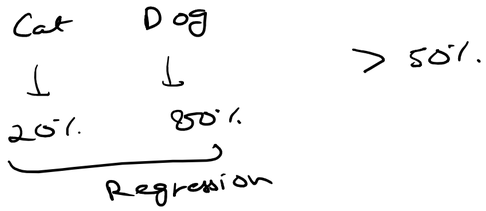


Predicting continuous values Eg. House price. Temperature Predict a number	Predicting fixed categories only. Eg. Pass/Fail Ratings given to uber driver

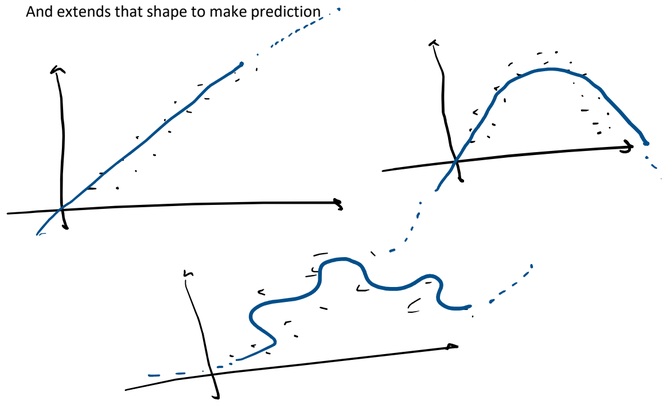
Which do you think happened first, Regression or Classification

Regression came into the world
Classification is actually -> Regression + A cutoff





Rough idea of Linear Regression, it tries to fit a shape to your data
And extends that shape to make prediction

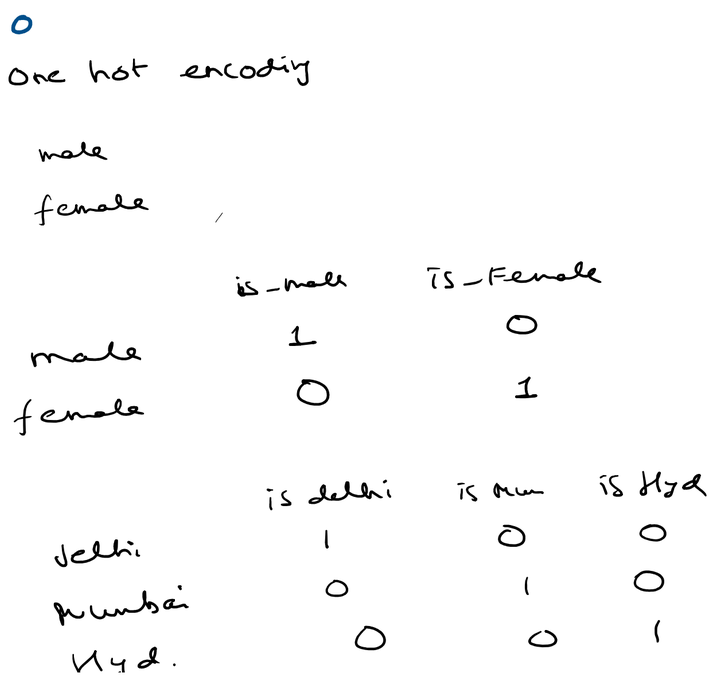


ML Pipeline:
1. Training -> We feed data into the model and it is trained to identify patterns
2. Inference -> Use the trained model for inference purposes.

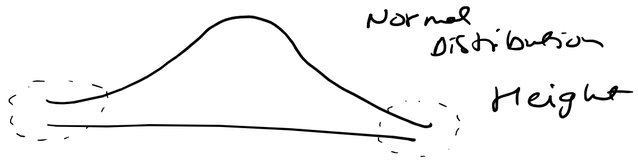
Steps need to take, before starting model training:

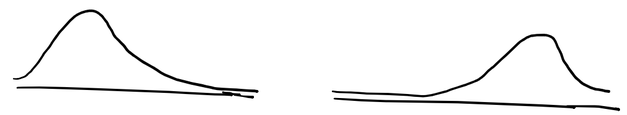
1. Data Preprocessing
    a. We can choose a great model with bad quality unclean data
    b. We can choose an okay okay model with great quality data

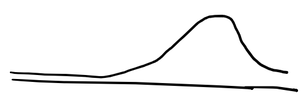
Data Preprocessing:



Hyd. 0 0 1







Doctor: Lazy → Not Sick

100 people

Town 90 healthy 10 sick

Not sick for all

$\frac{90}{100} \times 100 = 90\%$ Accuracy

Confusion Matrix	Actual Negative	Actual Positive
Predict Negative	90 (TN)	10 (FN)
Predict positive	0 (FP)	0 (TP)

90 -ve, 10 +ve

100 Actual -ve

Recall =

90%.

Recall : Out of total actual positives, how many our model was able to catch

$\frac{TP}{TP+FN} = 0$

Town → 100 → 90 Actual -ve
10 Actual +ve

Dr. Panic → Predicts +ve for all

	Actual -ve	Actual +ve
Predict -ve	0 TN	0 FN
Predict +ve	90 FP	10 TP

Recall = $\frac{TP}{TP+FN}$

$= \frac{10}{10+0}$

$= 1$

$= 100\%$

Acc. $\frac{TP+TN}{TP+FP+TN+FN}$

Precision = $\frac{TP}{TP+FP}$

$= \frac{10}{10+90}$

$= \frac{10}{100} = 0.1$

$= 10\%$

Out of total predicted positives, how many were correct

A policeman, stopping cars and checking them on basis of his suspicion.

Case 1. If he stops almost everybody!
Recall -> High
Precision -> Low

Case 2. If he stops almost nobody
Recall -> Low
Precision -> High

Town, 100 People
60 Not sick, 40 Sick

Actual -ve TN Actual +ve FN

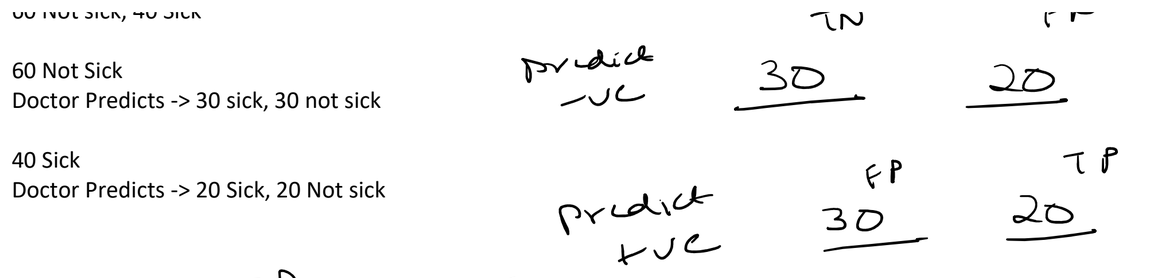
Predict

60 Not Sick

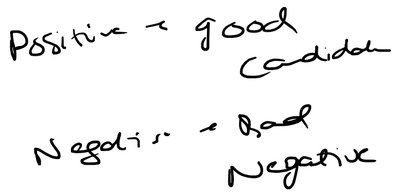
60 Not sick, 40 Sick

60 Not Sick
Doctor Predicts -> 30 sick, 30 not sick

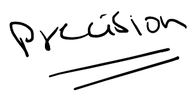
40 Sick
Doctor Predicts -> 20 Sick, 20 Not sick



1. HR is desperately trying to find a good candidate.
   The resume shortlist algorithm
   Should its precision be high or recall be high?



2. Conducting an interview costs 5000 rupees.
   Recall high or precision high?



We can combine precision and recall into one number, that is F1 Score

Can I take simple mean of precision + recall

F1 = (Precision + Recall) / 2

In simple mean, even if one of them is big, the whole mean increases

We need to take harmonic mean

F1 = (2*Precision*Recall) / (Precision + Recall)

IT will be maximum only when both are high.